

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T18 · Expanding Brackets

With Answers

Questions with full worked solutions

■ 42 pages

FAMILY MAP

Expanding Brackets

F01

Expand three or more linear factors

12 Q | 34 marks

F02

Expand two linear expressions

5 Q | 10 marks

F03

Distribute a single term

3 Q | 11 marks

Working Unit 19

4MA1-2H Q12

ORIGINAL QUESTION

3 marks

12 Expand and simplify $3x(2x+5)(7x-4)$

(Total for Question 12 is 3 marks)

Worked Solution 19

Expand three algebraic factors

UNIT 1

3 marks

First two brackets

$$(2x + 5)(7x - 4) = 14x^2 + 27x - 20.$$

Multiply by 3x

$$3x(14x^2 + 27x - 20) = 42x^3 + 81x^2 - 60x.$$

Final answer

$$42x^3 + 81x^2 - 60x$$

Question 02

4MA1-1H Q6

ORIGINAL QUESTION

2 marks

6 (a) Expand and simplify $(m + 5)(m - 8)$

.....
(2)

Worked Solution 02

Expand brackets

MODULAR UNIT 1

2 marks

Expand

$$(m + 5)(m - 8) = m^2 - 3m - 40.$$

Final answer

$$m^2 - 3m - 40$$

Question 28

4MA1-2H Q16 whole

ORIGINAL QUESTION UNIT 1

3 marks

16 Expand and simplify $(2x + 3)(x - 5)(x + 4)$

.....

Worked solution

METHOD / WORKING

MODULAR UNIT 1 • NOVEMBER 2023 DETAILED SOLUTIONS
Worked solution 28 MODULAR UNIT 1
Expand three brackets 3 marks
whole – Expand three brackets
Multiply two brackets
$(x - 5)(x + 4) = x^2 - x - 20.$
Multiply by the third bracket
$(2x + 3)(x^2 - x - 20) = 2x^3 - 2x^2 - 40x + 3x^2 - 3x - 60.$
Collect like terms
$2x^3 + x^2 - 43x - 60.$
Final answer
$2x^3 + x^2 - 43x - 60$

FINAL ANSWER

Question 22

4MA1-2H Q17

ORIGINAL QUESTION

3 marks

17 (a) Expand and simplify $(x + 6)(3x - 2)(x + 6)$

(3)

Worked Solution 22

Expand three brackets

MODULAR UNIT 1

3 marks

Square the repeated factor

$$(x + 6)^2 = x^2 + 12x + 36.$$

Multiply by the third factor

$$(x^2 + 12x + 36)(3x - 2) = 3x^3 + 34x^2 + 84x - 72.$$

Final answer

$$3x^3 + 34x^2 + 84x - 72$$

Original question 02

4MA1-1H Q3 M02

MODULAR UNIT 1

2 marks

(b) Expand and simplify $2n(4n + 3) + n(n - 4)$

.....
(2)

Worked solution 02

Expand and simplify

MODULAR UNIT 1

2 marks

M02 – Expand and simplify**Expand both products**

$$2n(4n + 3) + n(n - 4) = 8n^2 + 6n + n^2 - 4n.$$

Collect like terms

$$= 9n^2 + 2n.$$

Final answer

$$9n^2 + 2n$$

Original question 11

4MA1-1H Q15 M01

MODULAR UNIT 1

3 marks

15 Write $3x(2x - 1)(5x + 4)$ in the form $ax^3 + bx^2 + cx$ where a , b and c are integers.

.....

Worked solution 11

Expand three factors

MODULAR UNIT 1

3 marks

M01 – Expand three factors**Expand the two brackets**

$$(2x - 1)(5x + 4) = 10x^2 + 3x - 4.$$

Multiply by 3x

$$3x(10x^2 + 3x - 4) = 30x^3 + 9x^2 - 12x.$$

Read a,b,c

$$a = 30, b = 9, c = -12.$$

Final answer

$$30x^3 + 9x^2 - 12x$$

Original question 02

4MA1-1H Q5 Q5(a)

ORIGINAL QUESTION**2 marks**

5 (a) Expand and simplify $(n - 6)(n + 4)$

.....
(2)

Worked solution

Family: Expand two linear expressions • Variant: Standard double bracket

METHOD / WORKING

MODULAR UNIT 1 • JUNE 2022 DETAILED SOLUTIONS
Worked solution 02 MODULAR UNIT 1
Chapter: Expanding Brackets 2 marks
Q5(a) – Chapter: Expanding Brackets
Family: Expand two linear expressions • Variant: Standard double bracket
Method
$(n - 6)(n + 4) = n^2 + 4n - 6n - 24.$
Why: Distribute each term in the first bracket across the second.
Method
<i>Collecting like terms gives</i> $n^2 - 2n - 24.$
Why: Combine $4n$ and $-6n$.
Final answer
<i>expansion</i> : $n^2 - 2n - 24$

FINAL ANSWER

Original question 22

4MA1-2H Q14 Q14(a)

ORIGINAL QUESTION**3 marks**

- 14 (a) Expand and simplify $(5 - x)(2x + 3)(x + 4)$
Show your working clearly.

(3)

Worked solution

Family: Expand three or more linear factors • Variant: Three linear brackets

METHOD / WORKING

MODULAR UNIT 1 • JUNE 2022 DETAILED SOLUTIONS
Worked solution 22 MODULAR UNIT 1
Chapter: Expanding Brackets 3 marks
Q14(a) – Chapter: Expanding Brackets
Family: Expand three or more linear factors • Variant: Three linear brackets
Method
$(5 - x)(2x + 3) = -2x^2 + 7x + 15.$
Why: Expand and collect the first pair of factors.
Method
$(-2x^2 + 7x + 15)(x + 4) = -2x^3 - 8x^2 + 7x^2 + 28x + 15x + 60.$
Why: Distribute every term across the third factor .
Method
<i>The simplified result is</i> $-2x^3 - x^2 + 43x + 60.$
Why: Collect like powers of x.
Final answer
<i>expansion :</i> $-2x^3 - x^2 + 43x + 60$

FINAL ANSWER

Original question 10

4MA1-1H Q13 Q13(a)

ORIGINAL QUESTION

3 marks

13 (a) Expand and simplify $5x(x + 2)(3x - 4)$

3

(3)

Worked solution

Family: Expand three or more linear factors • Variant: Monomial times two linear brackets

METHOD / WORKING

MODULAR UNIT 1 • JANUARY 2022 DETAILED SOLUTIONS
Worked solution 10 MODULAR UNIT 1
Chapter: Expanding Brackets 3 marks
Q13(a) – Chapter: Expanding Brackets
Family: Expand three or more linear factors • Variant: Monomial times two linear brackets
Method
$5x(x + 2) = 5x^2 + 10x.$
Why: Multiply the first two factors.
Method
$(5x^2 + 10x)(3x - 4) = 15x^3 - 20x^2 + 30x^2 - 40x.$
Why: Multiply by the third factor .
Method
<i>Collecting terms gives</i> $15x^3 + 10x^2 - 40x.$
Why: Combine the two x^2 terms.
Final answer
<i>expanded expression</i> : $15x^3 + 10x^2 - 40x$

FINAL ANSWER

Original question 18

4MA1-2H Q1 Q1(a)

ORIGINAL QUESTION**2 marks**

- 1 (a) Expand and simplify $(y + 4)(2 - y)$

.....
(2)

Worked solution

Family: Expand two linear expressions • Variant: Opposite signs, repeated bracket or square

METHOD / WORKING

MODULAR UNIT 1 • JANUARY 2022 DETAILED SOLUTIONS
Worked solution 18 MODULAR UNIT 1
Chapter: Expanding Brackets 2 marks
Q1(a) – Chapter: Expanding Brackets
Family: Expand two linear expressions • Variant: Opposite signs, repeated bracket or square
Method
$(y + 4)(2 - y) = 2y + 8 - y^2 - 4y.$
Why: Expand both products.
Method
<i>Collecting terms gives</i> $8 - 2y - y^2.$
Why: Combine the y terms.
Final answer
<i>expanded expression</i> : $8 - 2y - y^2$

FINAL ANSWER

7 (a) Expand and simplify $(m - 8)(m + 5)$

(2)

Worked solution

January 2020 | Question 7 | 2 marks

Family: Expand two linear expressions | Variant: Standard double bracket

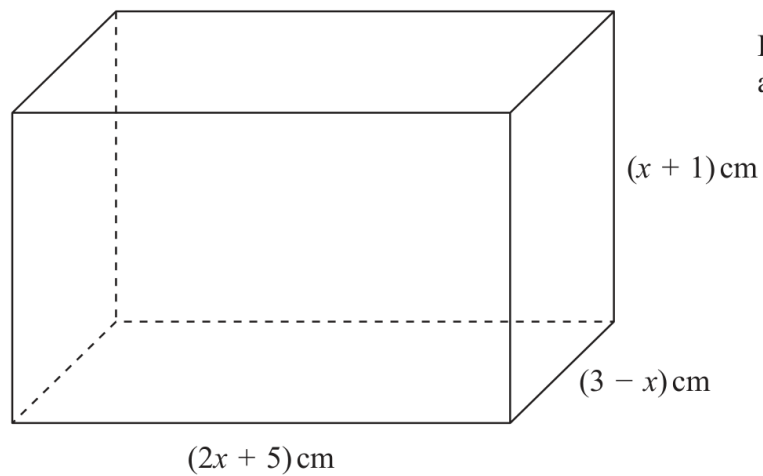
METHOD / WORKING

1. *Expand*: $m^2 + 5m - 8m - 40$.2. *Collect terms* to get $m^2 - 3m - 40$.

FINAL ANSWER

$$m^2 - 3m - 40$$

15



The diagram shows a cuboid of volume $V\text{cm}^3$

(a) Show that $V = 15 + 16x - x^2 - 2x^3$

(3)

Worked solution

January 2020 | Question 15 | 3 marks

Family: Expand three or more linear factors | Variant: Three linear brackets

METHOD / WORKING

1. $V = (2x+5)(x+1)(3-x)$.

2. $(2x + 5)(x + 1) = 2x^2 + 7x + 5$.

3. *Expand with* $(3 - x)$ to obtain $V = 15 + 16x - x^2 - 2x^3$.

FINAL ANSWER

$$V = 15 + 16x - x^2 - 2x^3$$

(b) Expand and simplify $2x(x - 5)(x - 3)$

(3)

Worked solution

January 2021 | Question 12 | 3 marks

Family: Expand three or more linear factors | Variant: Monomial times two linear brackets

METHOD / WORKING

1. *First* $(x - 5)(x - 3) = x^2 - 8x + 15$.

2. *Multiply by* $2x : 2x^3 - 16x^2 + 30x$.

3. Collect terms and state the expansion.

FINAL ANSWER

$$2x^3 - 16x^2 + 30x$$

5 (a) Expand and simplify $4x(2x + 5) - 3x(2x - 3)$

(2)

Worked solution

January 2021 | Question 5 | 2 marks

Family: Distribute a single term | Variant: Negative term or subtraction of a bracket

METHOD / WORKING

1. *Expand*: $4x(2x + 5) = 8x^2 + 20x$ and $-3x(2x - 3) = -6x^2 + 9x$.

2. *Collect terms* to obtain $2x^2 + 29x$.

FINAL ANSWER

$$2x^2 + 29x$$

5 (a) Expand and simplify $3x(2x + 3) - x(3x + 5)$

(2)

Worked solution

June 2021 | Question 5 | 2 marks

Family: Expand three or more linear factors | Variant: Monomial times two linear brackets

METHOD / WORKING

1. *Expand* : $3x(2x + 3) - x(3x + 5) = 6x^2 + 9x - 3x^2 - 5x$.

2. *Collectliketerms* to obtain $3x^2 + 4x$.

FINAL ANSWER

$$3x^2 + 4x$$

- 13** Expand and simplify $4x(3x + 1)(2x - 3)$
Show your working clearly.

Worked solution

June 2021 | Question 13 | 3 marks

Family: Expand three or more linear factors | Variant: Monomial times two linear brackets

METHOD / WORKING

1. Expand two factors, for example $(3x + 1)(2x - 3) = 6x^2 - 7x - 3$.

2. Multiply by $4x$: $24x^3 - 28x^2 - 12x$.

3. State the collected expression.

FINAL ANSWER

$$24x^3 - 28x^2 - 12x$$

(b) Expand and simplify $(x - 3)(x + 1)$

(2)

Worked solution

November 2021 | Question 1 | 2 marks

Family: Expand two linear expressions | Variant: Standard double bracket

METHOD / WORKING

1. (b) $(x - 3)(x + 1) = x^2 - 2x - 3.$

2. Answer: (a) e^6 , (b) $x^2 - 2x - 3$

FINAL ANSWER

$$(b)x^2 - 2x - 3$$

- (c) Expand and simplify $4x(x - 5)(2x + 3)$
Show your working clearly.

(3)

Worked solution

November 2021 | Question 12 | 3 marks

Family: Expand three or more linear factors | Variant: Monomial times two linear brackets

METHOD / WORKING

1. (c)

$$2. 4x(x - 5)(2x + 3) = 4x(2x^2 - 7x - 15)$$

$$3. = 8x^3 - 28x^2 - 60x$$

$$4. \text{ Answer: (a) } 16p^6q^8, \text{ (b) } \frac{17}{30x}, \text{ (c) } 8x^3 - 28x^2 - 60x$$

FINAL ANSWER

$$(c) 8x^3 - 28x^2 - 60x$$

- 13** Expand and simplify $3x(2x - 5)^2$
Show clear algebraic working.

(Total for Question 13 is 3 marks)

Answer reference | January 2023 | Question 13

Family: Distribute a single term • Variant: Positive monomial over a bracket

13	$3x(2x-5) = 6x^2 - 15x$ or		3	M1 for multiplying $3x$ by $(2x-5)$ with both terms correct or
	$(2x-5)^2 = 4x^2 - 10x - 10x + 25$ or			for multiplying $(2x-5)$ by $(2x-5)$ with 3 out of 4 terms correct or
	$(2x-5)^2 = 4x^2 - 20x + 25$			for multiplying $(2x-5)$ by $(2x-5)$ and getting $4x^2 - 20x \dots$ or $\dots -20x + 25$ (not for $4x^2 + 25$)
	$(6x^2 - 15x)(2x-5) = 12x^3 - 30x^2 - 30x^2 + 75x$ oe or			M1ft (dep) for multiplying the product of $3x$ and $(2x-5)$ by $(2x-5)$ with 3 out of 4 terms correct or
	$(6x^2 - 15x)(2x-5) = 12x^3 - 60x^2 + 75x$ oe or			for multiplying the product of $3x$ and $(2x-5)$ by $(2x-5)$ and getting $12x^3 - 60x^2 \dots$ or $\dots -60x^2 + 75x$
	$3x(4x^2 - 10x - 10x + 25) = 12x^3 - 30x^2 - 30x^2 + 75x$ oe or			for multiplying the product of $(2x-5)$ and $(2x-5)$ by $3x$ with 3 out of 4 terms correct or
	$3x(4x^2 - 20x + 25) = 12x^3 - 60x^2 + 75x$			for multiplying the product of $(2x-5)$ and $(2x-5)$ by $3x$ with 2 out of 3 terms correct or
				Expansion in one stage will lead to $12x^3 - 30x^2 - 30x^2 + 75x$ without firstly expanding two factors – award M2 for 3 out of 4 terms correct M1 for 2 out of 4 terms correct
	<i>Working required</i>	$12x^3 - 60x^2 + 75x$		A1 dep on M1
				Total 3 marks

Worked solution

January 2023 | Question 13 | 3 marks

Family: Distribute a single term • Variant: Positive monomial over a bracket

Official final mark-scheme pages are embedded immediately after this question.

14 (a) Expand and simplify $7x(3x + 2)(2x - 5)$

(3)

(b) Solve $\frac{9}{2y} + \frac{5}{7} = 5$

Show clear algebraic working.

$y =$
(3)

(Total for Question 14 is 6 marks)

Worked solution

November 2025 | Question 14 | 6 marks

Family: Distribute a single term | Variant: Positive monomial over a bracket

METHOD / WORKING

Worked solution
November 2025 Question 14 6 marks
Family: Distribute a single term Variant: Positive monomial over a bracket
Solution
(a)
$7x(3x + 2)(2x - 5) = 7x(6x^2 - 11x - 10)$
$= 42x^3 - 77x^2 - 70x$
(b)
$\frac{9}{2}y + \frac{5}{7} = 5$
$\frac{9}{2}y = (5 - \frac{5}{7})$
$63 = 60y$
Answer: (a) $42x^3 - 77x^2 - 70x$, (b) $y = \frac{21}{20}$
PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER